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Summary

Senior software engineer, 8+ years across high-throughput data infrastructure and developer tooling in Haskell, by applying programming-language theory to production systems at scale.

Employment

Senior Software Engineer, Core | **Scrive** | **08/2025 — Present**

- Designed and implemented a high-stakes migration, integrating a legacy system serving the main customer base into new AI-based workloads.
- Improved existing CI processes for the main Haskell codebase, reducing integration lead times by ~70%, and increasing deployment velocity.
- Designed and implemented automated methods for internal data traces, easing complex debug scenarios.
- Improved developer experience by facilitating local environment bootstrapping of services.

Software Engineer, Infra | **Channable** | **02/2021 — 08/2025**

- Refactored infrastructure responsible for importing terabytes of data from external services per day, improving debuggability and observability.
- Designed and implemented AI-assisted categorization using state-of-the-art techniques for mass text classification, improving the existing model performance by 3x.
- Led integration of type-safe user-defined computations into a high-throughput compute pipeline, applying programming-language theory to guarantee safety without sacrificing performance.
- Redesigned core scheduling infrastructure after analyzing its expressivity and usability limits, driving adoption through well-researched proposals and cutting compute overhead by multiple CPU-weeks per day.

Software Engineer | **Cargowatch** | **02/2018 — 12/2020**

- Implemented a specialized web portal for customer support and designed a DSL for invoicing specification.
- Algorithmically improved the existing automatic invoicing process.

Education

MSc. BSc. Computing Science | **Utrecht University** | **2015 — 2020**

- Thesis: Formalized Correctness Proofs of Automatic Differentiation in Coq. Proof using logical relations accompanied by simple language representations and denotational semantics.

Projects

Maintainer, Haskell Language Server | https://github.com/haskell/haskell-language-server

- Sped up raw indexing speed of GHC-emitted interface files by ~50% through optimizing internal SQLite usage.
- Implemented a lsp-recorder that can be used to debug realistic editing workloads on arbitrary LSP servers.
- Implemented `hls-export-plugin` for mutating export lists intelligently.

Helium

- Contributed to the Helium Haskell compiler by implementing and integrating missing Haskell2010 features.

Programming Languages and Technologies

- **Programming Languages**: Haskell, Python, Nix, SQL
- **Technologies**: Git, PostgreSQL, SQLite, Docker
- **Languages**: Dutch, English